SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED
CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct
I Thamaraiselvi S (192511042) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Thamaraiselvi S

Assessment Tasks
Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Assessment Tasks – Short Answer Test (SET 1)

1. Define Cloud Computing and Explain Any Four Characteristics That Make It Different from Traditional On-Premise Computing

Definition

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the Internet ("the cloud"). Instead of purchasing and maintaining physical hardware, users can access these resources whenever needed and pay only for what they use.

Characteristics of Cloud Computing

1. On-Demand Self-Service

- Users can access computing resources without contacting the service provider.
- Resources such as virtual machines and storage are available instantly.

Example: Creating an AWS EC2 virtual machine in a few minutes.

2. Broad Network Access

- Cloud services are available over the Internet.
- Users can access them using laptops, smartphones, tablets, or desktops from anywhere.

Example: Accessing Google Drive from any device.

3. Resource Pooling

- Cloud providers share computing resources among multiple customers using virtualization.
- Resources are dynamically assigned according to demand.

Example: Multiple organisations using the same physical server securely.

4. Rapid Elasticity

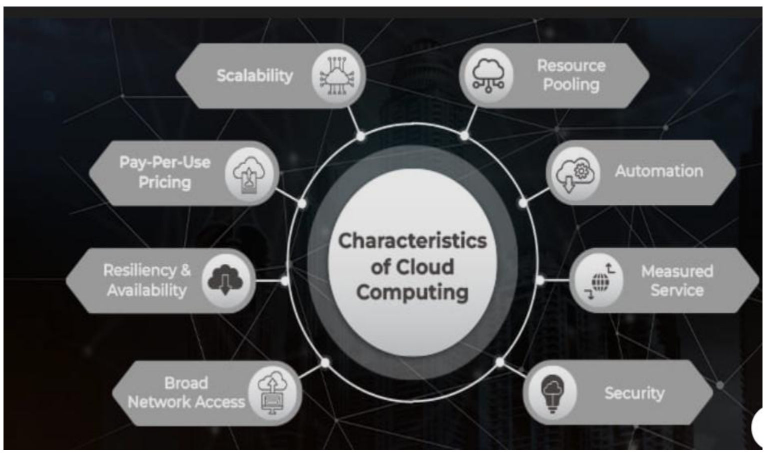
- Resources can automatically increase or decrease based on workload.
- This avoids overloading systems and reduces costs.

Example: Shopping websites increasing servers during festival sales.

Table 1: Difference from Traditional On-Premise Computing

Cloud Computing	Traditional On-Premise Computing
Internet-based	Local infrastructure
Pay-as-you-use	High initial investment
Easily scalable	Difficult to scale
Managed by cloud provider	Managed by organisation

Fig 1: Characteristics of cloud computing



2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution

Evolution Timeline

Stage 1: Mainframe Computing (1960s)

- Large centralized computers served many users.
- Users accessed systems through terminals.

Stage 2: Personal Computers (1980s)

- Individual users owned their own computers.
- Processing shifted from central systems to local machines.

Stage 3: Client-Server Computing (1990s)

- Clients requested services from dedicated servers.
- Databases and applications were stored centrally.

Stage 4: Internet and Web Applications (2000s)

- High-speed Internet enabled online applications.
- Services became accessible worldwide.

Stage 5: Virtualization

- Multiple virtual machines ran on one physical server.
- Increased hardware utilisation and flexibility.

Stage 6: Cloud Computing

- Providers such as AWS, Microsoft Azure and Google Cloud offered computing resources over the Internet.
- Users could rent resources instead of buying hardware.

Evolution Flow

Mainframe

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Personal Computers

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Client-Server

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Internet Applications

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Virtualization

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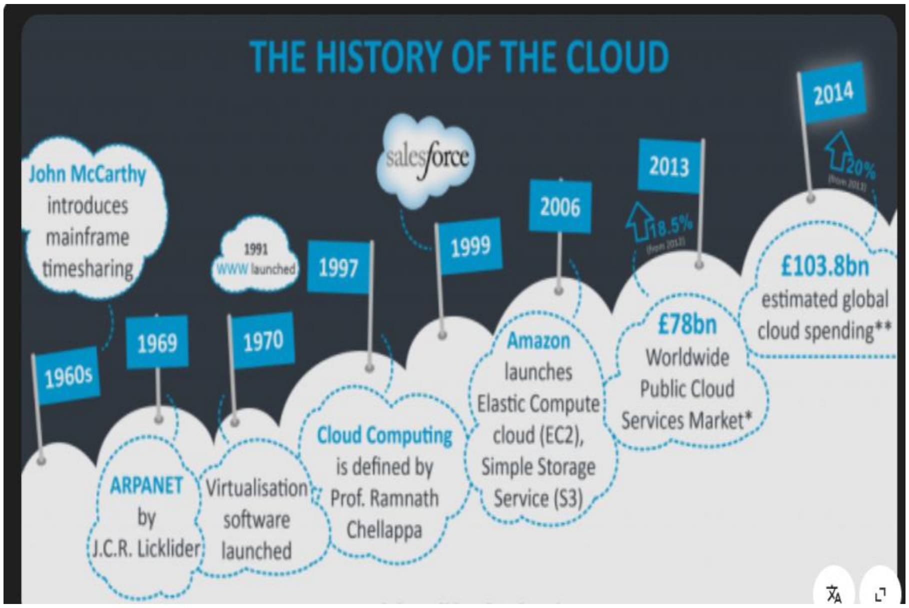
Cloud Computing

Fig 2: History of the cloud



3. Explain the Role of Server Virtualization in Cloud Computing

What is Server Virtualization?

Server virtualization is the process of dividing one physical server into multiple virtual servers using a hypervisor.

Each virtual server:

- Has its own operating system
- Runs independently
- Shares the physical hardware

Role in Cloud Computing

- Improves hardware utilisation.
- Reduces infrastructure cost.
- Enables quick deployment.
- Supports scalability.
- Provides isolation between users.
- Simplifies backup and disaster recovery.

Table 2:Virtualization vs Cloud Computing

Virtualization	Cloud Computing
Technology	Service model
Creates virtual machines	Delivers IT services
Uses hypervisors	Uses virtualization plus automation
Runs on one server	Accessible through Internet
Foundation of cloud	Complete cloud platform

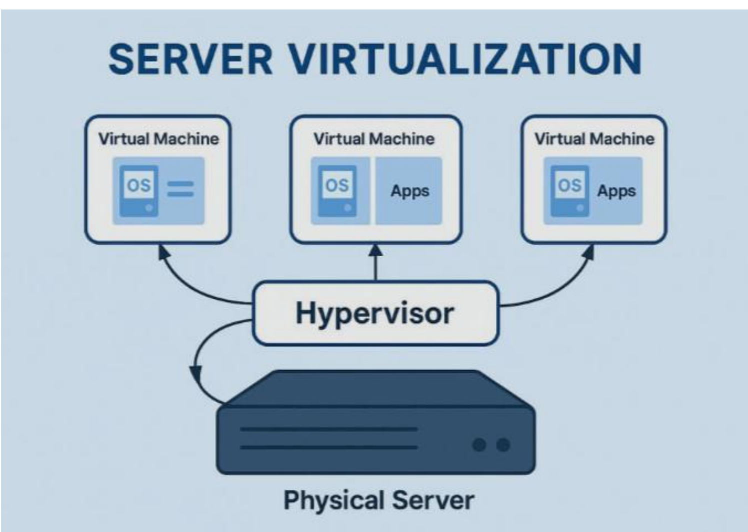
Example

One physical server hosting:

- VM 1 – Web Server
- VM 2 – Database
- VM 3 – Mail Server

All operate independently.

Fig 3: Server Virtualization



4. Discuss the Role of Data Centers in Delivering Cloud Services

What is a Data Center?

A data center is a secure facility containing servers, storage devices, networking equipment and cooling systems that provide cloud services.

Role of Data Centers

1. Storage

Stores user files, databases and backups.

2. Computing Power

Runs applications, websites and cloud services.

3. Networking

Provides high-speed Internet connectivity.

4. Security

Protects systems using:

- Firewalls
- Encryption
- CCTV
- Biometric access

5. Backup and Disaster Recovery

Maintains multiple copies of data for recovery.

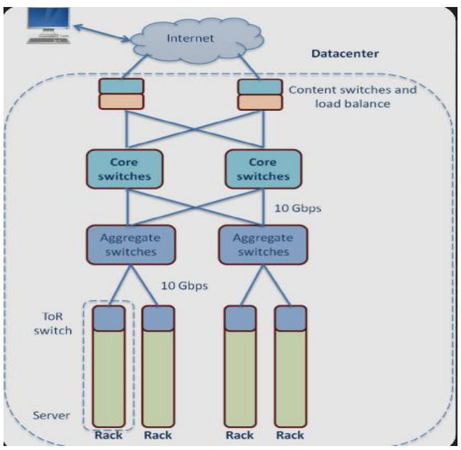
6. High Availability

Uses redundant servers and power supplies to ensure continuous service.

Components of a Data Center

- Servers
- Storage devices
- Routers and switches
- Power supply
- Cooling system
- Firewalls
- Backup systems

Fig 4: Data center



5. Differentiate IaaS, PaaS, SaaS and XaaS with One Suitable Real-World Example

Comparison Table

Service Model	**Full Form**	**Description**	**User Manages**	**Example**
IaaS	Infrastructure as a Service	Provides virtual machines, storage and networking	Operating system, applications and data	Amazon EC2
PaaS	Platform as a Service	Provides a platform to build and deploy applications	Applications and data	Google App Engine
SaaS	Software as a Service	Ready-to-use software over the Internet	Only user settings and data	Microsoft 365
XaaS	Anything as a	Any IT service delivered	Depends on the service	Dropbox (Storage

Service Model	Full Form	Description	User Manages	Example
	Service	through the cloud		as a Service)

Brief Explanation

IaaS (Infrastructure as a Service)

- Provides virtual infrastructure.
- Users install operating systems and applications.

Example: Amazon EC2

PaaS (Platform as a Service)

- Provides development tools and runtime environments.
- Developers focus on coding without managing infrastructure.

Example: Google App Engine

SaaS (Software as a Service)

- Complete software delivered through a web browser.
- No installation or maintenance required.

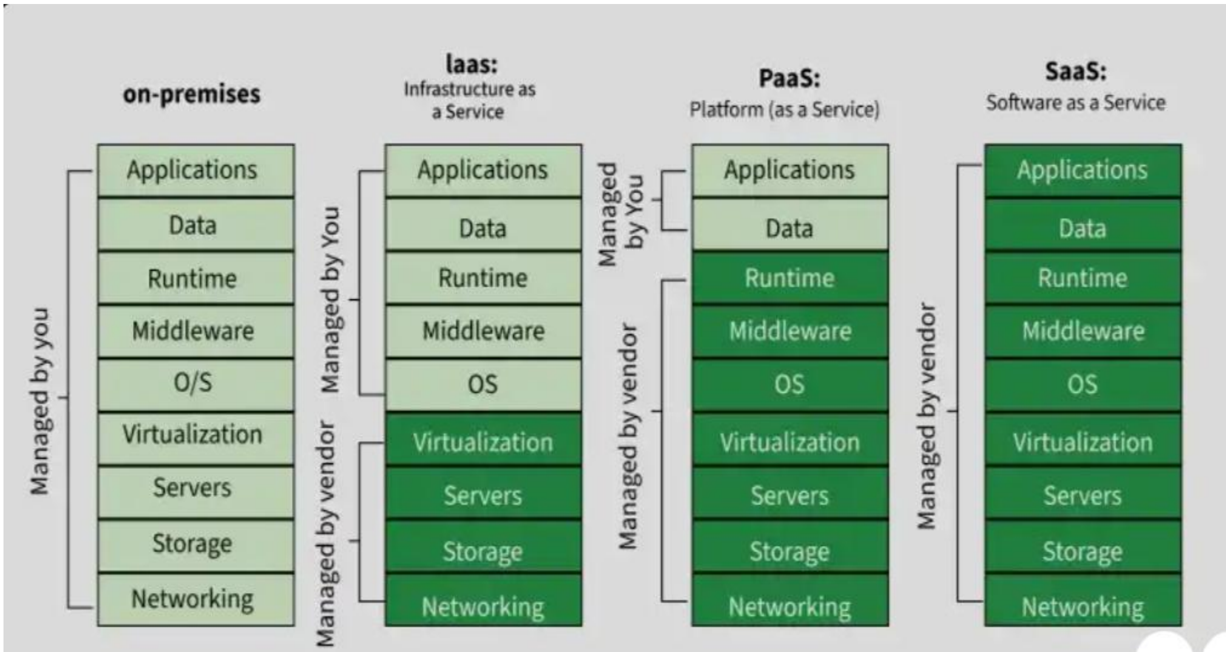
Example: Microsoft 365

XaaS (Anything as a Service)

- Refers to any cloud-delivered service such as Storage, Database, Security or AI.

Example: Dropbox (Storage as a Service)

Fig 5: IaaS, PaaS, SaaS and XaaS



Conclusion

Cloud computing provides flexible, scalable and cost-effective IT resources over the Internet. Its key characteristics—on-demand self-service, broad network access, resource pooling and rapid elasticity—make it different from traditional on-premise computing. The evolution from mainframes to virtualization enabled today's cloud platforms, while data centres and server virtualization form the backbone of cloud services. Service models such as **IaaS, PaaS, SaaS and XaaS** allow users to choose the level of control and management that best suits their needs.